

ZEBRA RFID AI DESIGN DOCUMENT

Zebra RFID AI Technical Design

Architecture, Connection Lifecycle, and Gemini AI Assistant Integration

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System architecture, connection state flow, activity lifecycle handling, UI overview, and implementation highlights for the Zebra RFID AI Android application.

Contents

1. System Architecture & Connection State Flow 2

2. Sequence Diagram: Connection & Lifecycle Flow 2

3. UI Screenshots & Visual Interface 2

4. Code Implementation Highlights 3

This document outlines the architecture, connection flows, activity lifecycle support (suspend & resume), inventory operations, and **Gemini 3.8 AI** assistant integration for the **Zebra RFID AI** Android application.

1. System Architecture & Connection State Flow

All Zebra RFID SDK operations are serialized on a single-threaded background executor (ExecutorService) in `RfidHandler.kt` to guarantee thread safety and maintain a smooth 60fps Jetpack Compose UI.

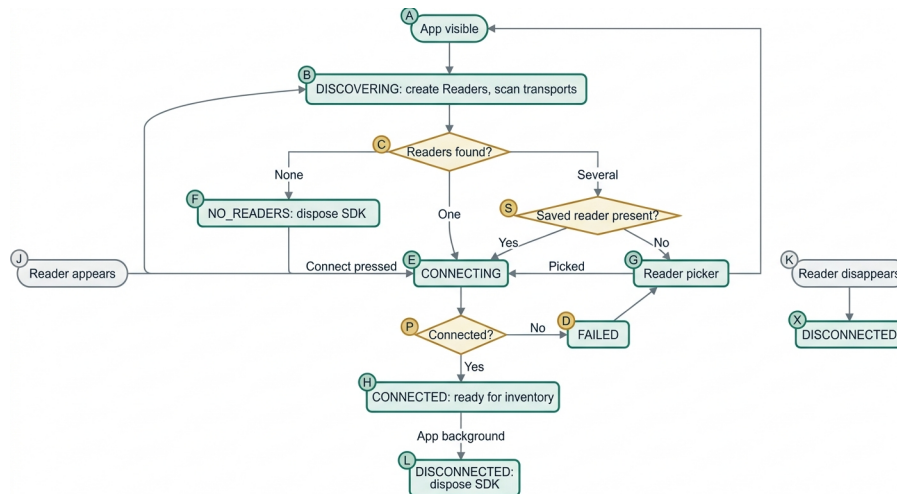


Figure 1: System Architecture & Connection State Flow

2. Sequence Diagram: Connection & Lifecycle Flow

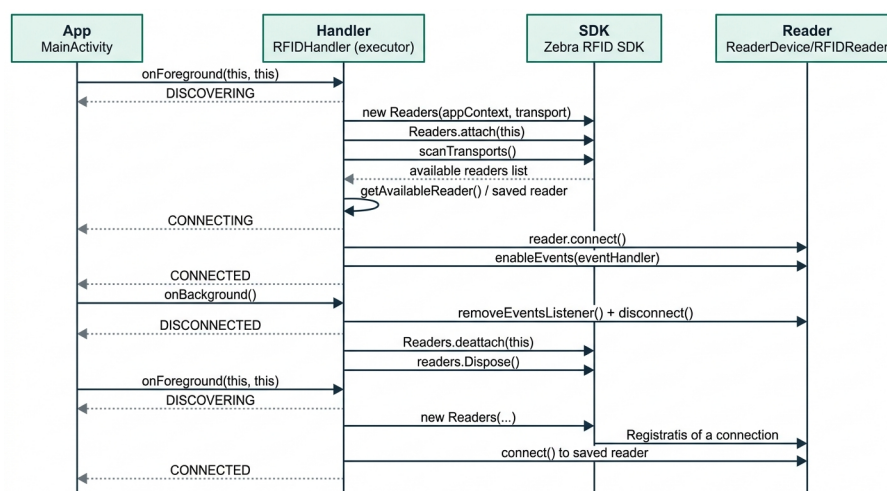


Figure 2: Connection & Lifecycle Sequence

3. UI Screenshots & Visual Interface

- **RFID Inventory Scanner:** real-time connection status, unique tag counts, total reads, EPC filter search bar, and inventory start/stop controls.

- **Gemini AI Assistant:** preset prompt chips for Zebra RFID SDK code snippets and inventory analysis.
- **Gemini AI Code Response:** Gemini 3.8 AI generated code explanation with local caching (↗ Cached) and force-refresh support.

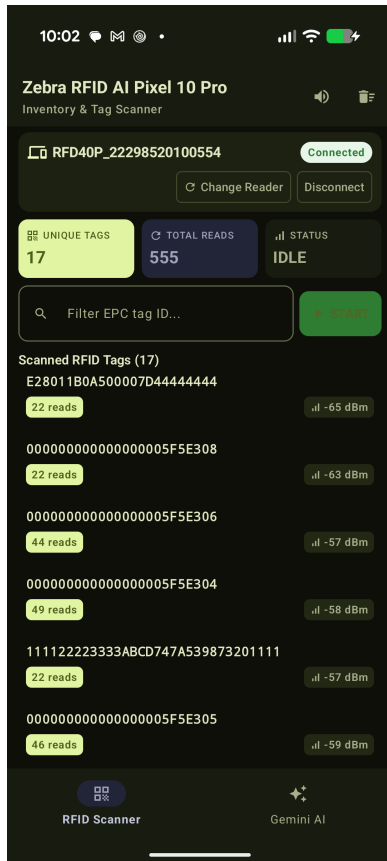


Figure 3: RFID Inventory Scanner

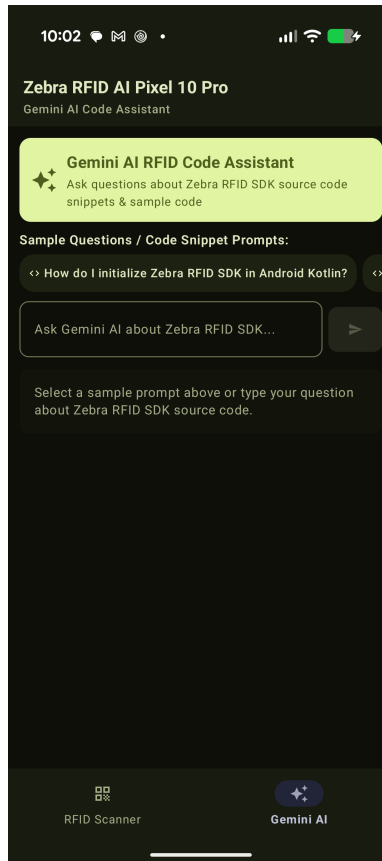


Figure 4: Gemini AI Assistant

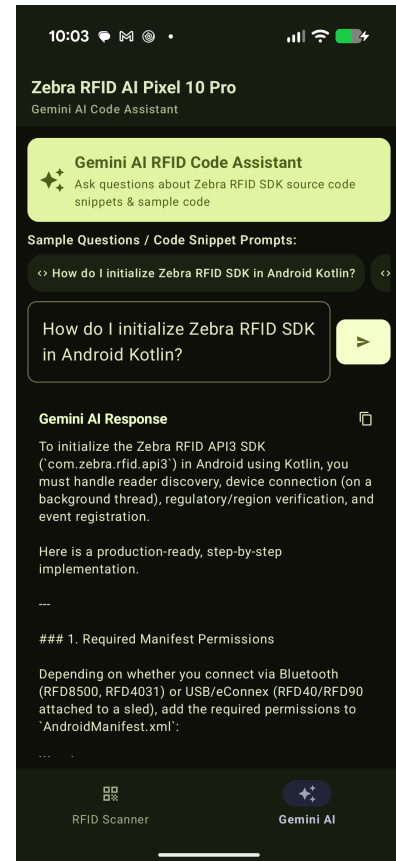


Figure 5: Gemini AI Response

4. Code Implementation Highlights

A. RFID Initialization, Discovery & Region Setup (RfidUtility.kt)

```
object RfidUtility {
    const val DEFAULT_READER_NAME = "RFID-TC701"

    @Throws(InvalidUsageException::class)
    fun scanTransports(readers: Readers): ArrayList<ReaderDevice> {
        for (candidate in ENUM_TRANSPORT.values()) {
            if (candidate == ENUM_TRANSPORT.ALL) continue
            try {
                readers.setTransport(candidate)
                val found = readers.GetAvailableRFIDReaderList()
                if (found != null && found.isNotEmpty()) return found
            } catch (_: InvalidUsageException) {}
        }
        return ArrayList()
    }
}
```

```

fun configureRegion(reader: RFIDReader, regionCode: String = "USA"): String? {
    try {
        val regCfg = reader.Config.regulatoryConfig
        val supportedRegions = reader.ReaderCapabilities.SupportedRegions
        for (i in 0 until supportedRegions.length()) {
            val regionInfo = supportedRegions.getRegionInfo(i)
            if (regionCode.equals(regionInfo.regionCode, ignoreCase = true)) {
                regCfg.region = regionInfo.regionCode
                regCfg.setIsHoppingOn(regionInfo.isHoppingConfigurable)
                regCfg.setEnabledChannels(regionInfo.supportedChannels)
                regCfg.setStandardName(regionInfo.name)
                reader.Config.regulatoryConfig = regCfg
                return null
            }
        }
        return "Region $regionCode not supported"
    } catch (e: Exception) {
        return e.message
    }
}

```

B. Gemini 3.8 AI Assistant with Local Caching (RfidAiViewModel.kt)

```

class RfidAiViewModel(application: Application) : AndroidViewModel(application) {
    private val generativeModel = Firebase.ai.generativeModel("gemini-3.8-flash")
    private val sharedPreferences = application.getSharedPreferences("gemini_cache",
        ↪ Context.MODE_PRIVATE)
    private val responseCache = ConcurrentHashMap<String, String>()

    fun askQuestion(userPrompt: String, scannedTags: List<TagItem> = emptyList(),
        ↪ forceRefresh: Boolean = false) {
        val cacheKey = userPrompt.trim().lowercase()
        if (!forceRefresh && responseCache.containsKey(cacheKey)) {
            _uiState.value = RfidAiUiState.Success(responseCache[cacheKey]!!, isCached =
            ↪ true)
            return
        }
        viewModelScope.launch(Dispatchers.IO) {
            val response = generativeModel.generateContent(content { text(userPrompt) })
            response.text?.let { output ->
                responseCache[cacheKey] = output
                sharedPreferences.edit().putString(cacheKey, output).apply()
                _uiState.value = RfidAiUiState.Success(output, isCached = false)
            }
        }
    }
}

```